

Exploratory Data Analysis & Business Intelligence on Retail Sales Data

Submitted By:
Karri Jyothi Archini
Internship Task 2

1. Objective:

The objective of this project is to perform Exploratory Data Analysis (EDA) on a retail sales dataset to identify patterns, trends, and business insights using Python and SQL. The project also aims to present the findings through visualizations and a dashboard.

2. Dataset Description :

The Superstore dataset contains information about customer orders, sales, profit, discount, quantity, product categories, and regions. It is used to analyze business performance and customer purchasing behavior. It contains 5 rows and 21 columns. Each row contains important columns such as Sales, Profit, Discount, Category, Region, and Quantity.

3. Descriptive Statistics :

Descriptive statistics were used to summarize the numerical features of the dataset. Metrics such as mean, standard deviation, minimum, and maximum values were analyzed.

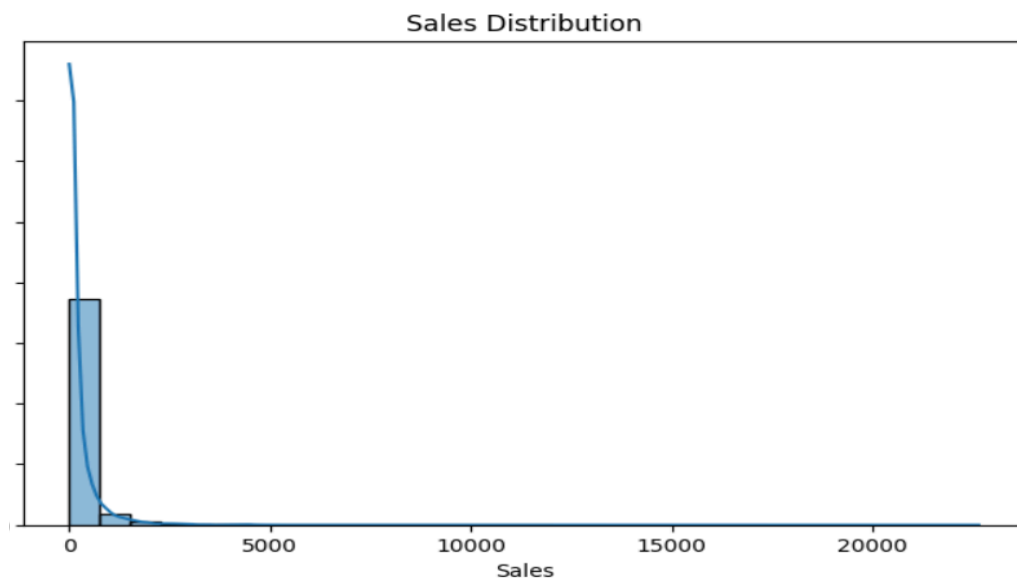
```
: print(df.describe())
```

	Row ID	Postal Code	Sales	Quantity	Discount \
count	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000
mean	4997.500000	55190.379428	229.858001	3.789574	0.156203
std	2885.163629	32063.693350	623.245101	2.225110	0.206452
min	1.000000	1040.000000	0.444000	1.000000	0.000000
25%	2499.250000	23223.000000	17.280000	2.000000	0.000000
50%	4997.500000	56430.500000	54.490000	3.000000	0.200000
75%	7495.750000	90008.000000	209.940000	5.000000	0.200000
max	9994.000000	99301.000000	22638.480000	14.000000	0.800000

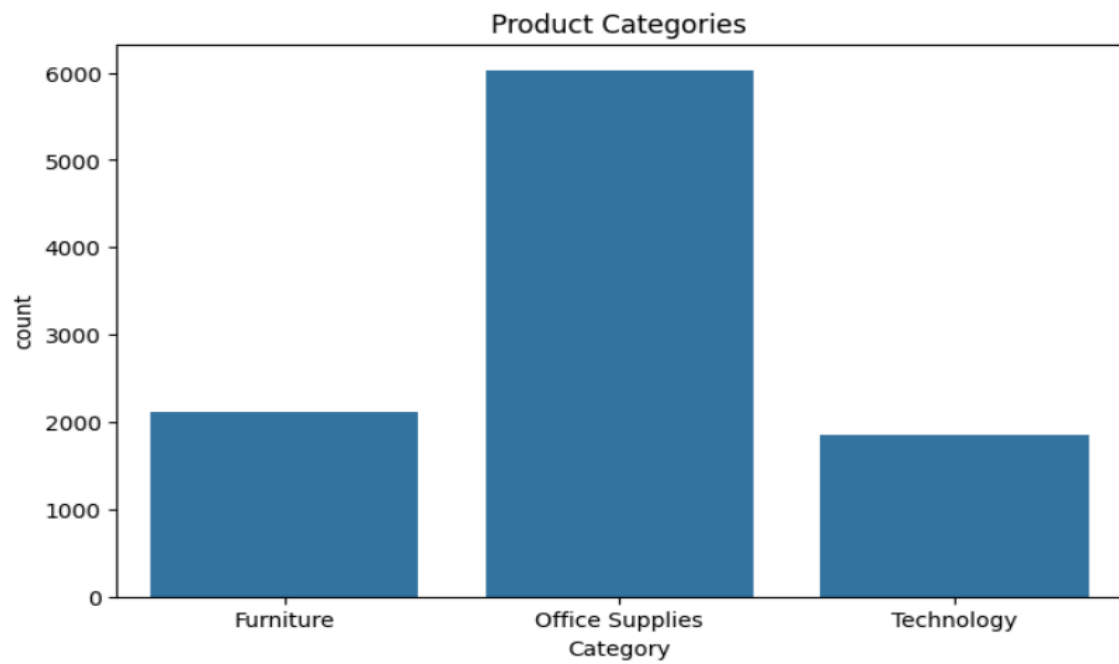
	Profit
count	9994.000000
mean	28.656896
std	234.260108
min	-6599.978000
25%	1.728750
50%	8.666500
75%	29.364000
max	8399.976000

4. Univariate Analysis :

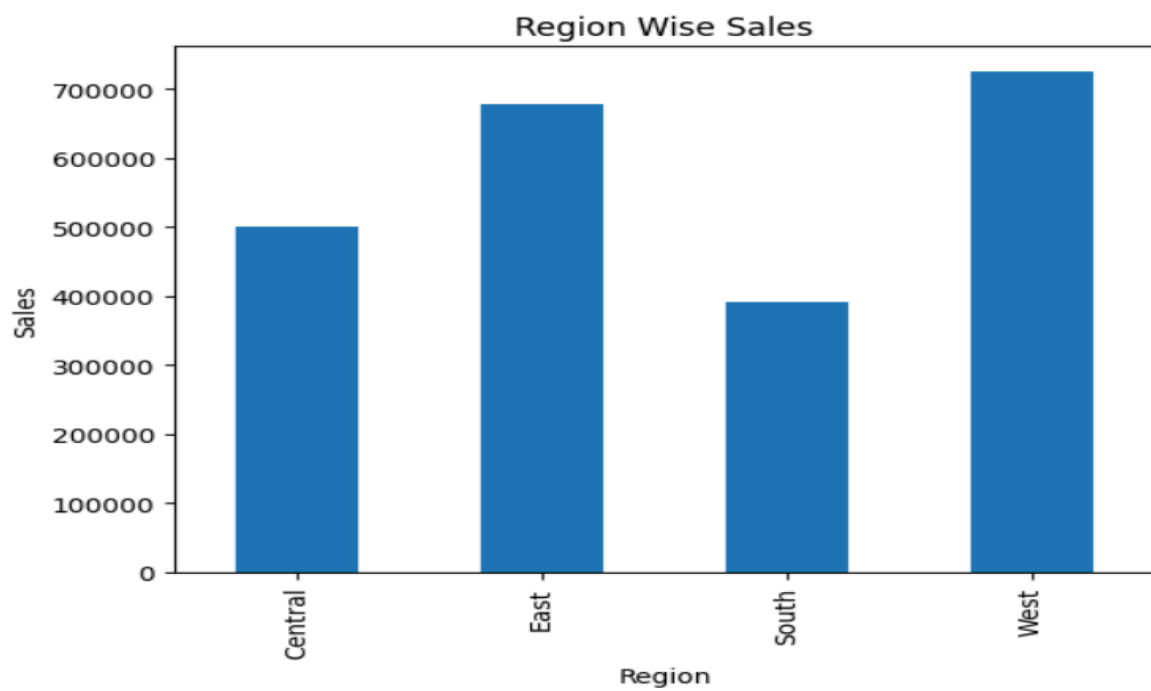
- Sales Distribution Histogram



- Category Count Plot



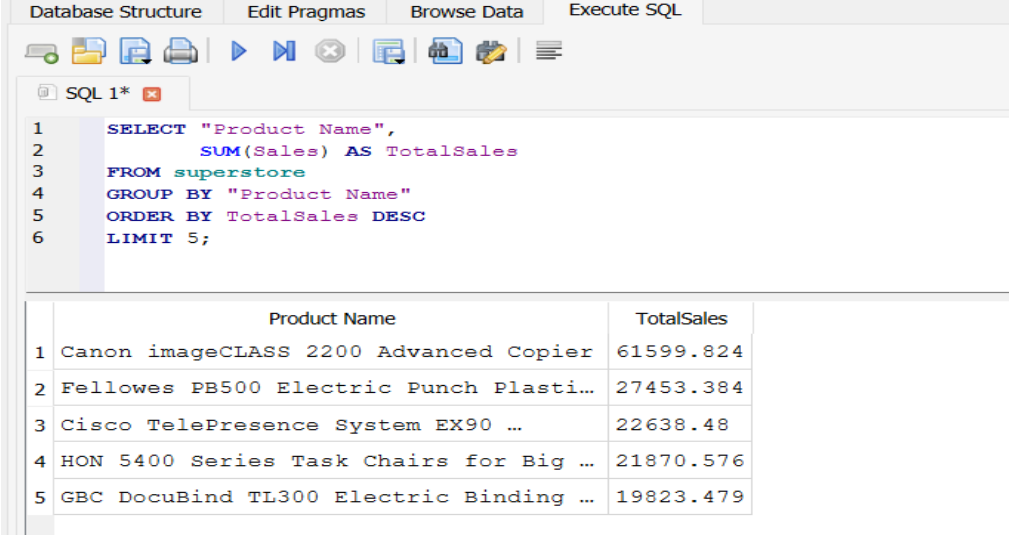
- Region-wise Sales Chart



5. SQL Business Analysis :

- Business Question 1: Which are the Top 5 Products by Sales?

```
SELECT "Product Name",
       SUM(Sales) AS TotalSales
FROM superstore
GROUP BY "Product Name"
ORDER BY TotalSales DESC
LIMIT 5;
```



The screenshot shows a SQL IDE interface with tabs for Database Structure, Edit Pragmas, Browse Data, and Execute SQL. The SQL editor contains the following query:

```
1 SELECT "Product Name",
2       SUM(Sales) AS TotalSales
3 FROM superstore
4 GROUP BY "Product Name"
5 ORDER BY TotalSales DESC
6 LIMIT 5;
```

Below the editor, the results of the query are displayed in a table with two columns: Product Name and TotalSales.

	Product Name	TotalSales
1	Canon imageCLASS 2200 Advanced Copier	61599.824
2	Fellowes PB500 Electric Punch Plasti...	27453.384
3	Cisco TelePresence System EX90 ...	22638.48
4	HON 5400 Series Task Chairs for Big ...	21870.576
5	GBC DocuBind TL300 Electric Binding ...	19823.479

- Business Question 2: Which Category Earned the Highest Profit?

```
SELECT Category,
       SUM(Profit) AS TotalProfit
FROM superstore
GROUP BY Category
ORDER BY TotalProfit DESC;
```

Database Structure

Edit Pragmas

Browse Data

Execute SQL



SQL 1* 

1

2

3

4

5

6

7

SELECT

Category,

SUM(Profit) AS TotalProfit

FROM superstore

GROUP BY Category

ORDER BY TotalProfit DESC;

	Category	TotalProfit
1	Technology	145454.9481
2	Office Supplies	122490.8008
3	Furniture	18451.2728

- Business Question 3: Which Region Generates the Highest Sales?

```
SELECT Region,
       SUM(Sales) AS Revenue
FROM superstore
GROUP BY Region
ORDER BY Revenue DESC;
```

Database Structure

Edit Pragmas

Browse Data

Execute SQL



SQL 1* 

1

2

3

4

5

6

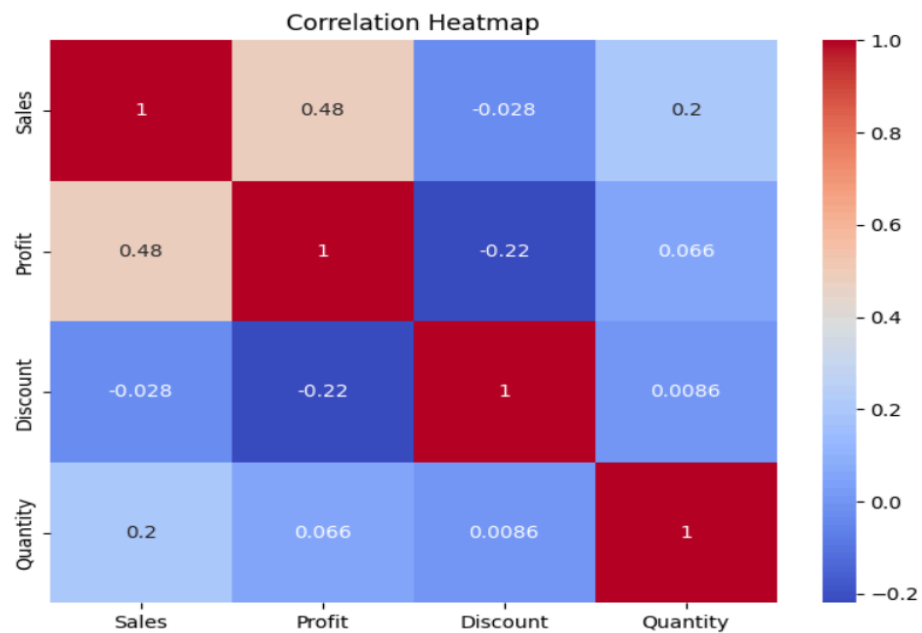
```
SELECT Region,
      SUM(Sales) AS Revenue
FROM superstore
GROUP BY Region
ORDER BY Revenue DESC;
```

	Region	Revenue
1	West	725457.8245
2	East	678781.24
3	Central	501239.8908
4	South	391721.905

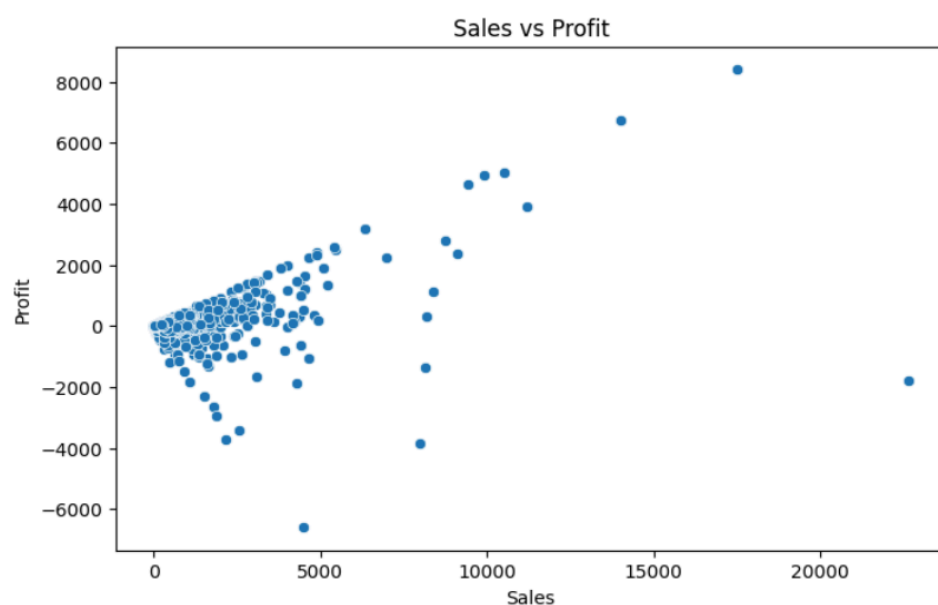
6. Correlation & Multivariate Analysis :

Correlation analysis showed a positive relationship between Sales and Profit. Discount showed a negative relationship with Profit, indicating that excessive discounts can reduce profitability.

- Correlation Heatmap

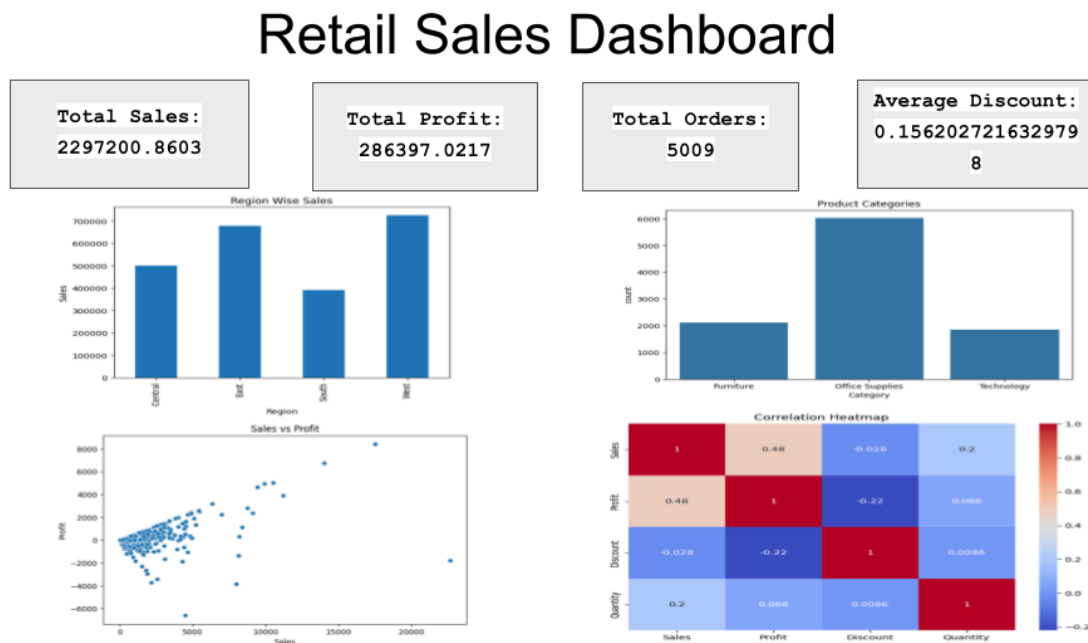


- Sales vs Profit Scatter Plot



7. Dashboard :

A static dashboard was created to summarize key performance indicators and important visualizations.



8. Key Insights :

- Technology category generated the highest profit.
- West region contributed the highest sales.
- Sales and Profit showed a positive correlation.
- Higher discounts often reduced profitability.
- Office Supplies had the highest number of transactions.
- A small number of products generated a large share of revenue.

9. Conclusion :

This project successfully analyzed retail sales data using EDA and SQL techniques. The findings revealed important business trends and provided actionable insights that can support decision-making.

